Unit 5

Campbell Biology, AP Edition, 12e

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The Evolutionary History of Biological Diversity

Sallie (Penny) Chisholm

Sallie (Penny) Chisholm, Institute Professor in Civil and Environmental Engineering at the Massachusetts Institute of Technology, has received the 2011 National Medal of Science, the 2019 Crafoord Prize in Biosciences, and many other prestigious awards. Dr. Chisholm received a B.A. in Biology and Chemistry from Skidmore College and a Ph.D. in Biology from the State University of New York, Albany. Elected to the U.S. National Academy of Sciences in 2003, Dr. Chisholm has a long-standing interest in the essential roles that photosynthetic microbes play in marine ecosystems. Currently, she and her students are studying these organisms from the genome to global scale.

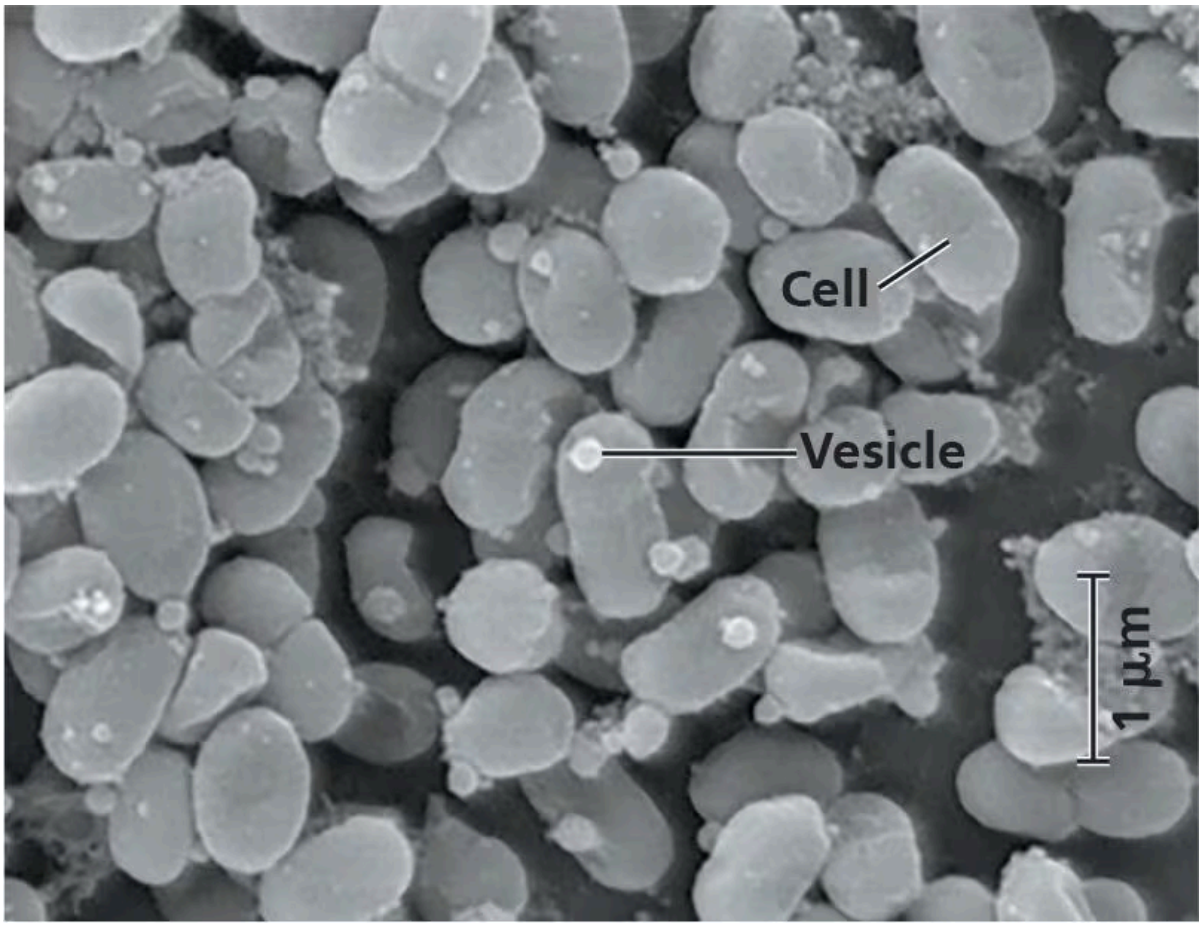
An Interview with: Penny Chisholm

How did you first become interested in science?

I grew up in Michigan's Upper Peninsula, in the sticks. Growing up, my family was all about business; I wasn't from an academic background. So, when I started studying biology in college, I was kind of surprised to find out that I liked science as a way of knowing. I had never been exposed to doing experiments and discovering new things on your own. I had the chance to do an independent research project on a local lake to measure its manganese budget. We measured the movement of manganese into and out of the lake over the course of a year, drilling holes in the ice to sample in winter. All of this was really cool and got me excited about the dynamics of aquatic ecosystems. Once I looked under the microscope and saw a world that I didn't even know existed—the microbes living in lakes—there was no turning back.

Much of your research is on the bacterium *Prochlorococcus*. Tell us about this organism.

Prochlorococcus is the smallest photosynthetic cell on the planet and the most abundant—it is less than 1/100th the width of a human hair, and there are over a billion billion billion of these tiny cells in the ocean! *Prochlorococcus* feeds small protists, which in turn are eaten by larger marine organisms, and thus they play a key role in ocean ecosystems. And when *Prochlorococcus* cells are infected by viruses, they break apart and their organic compounds serve as a food source for other bacteria in the ocean. Despite these organisms' importance, we didn't even know they existed when I was a student. And then, in the mid-1980s, I was fortunate enough to be involved in their discovery. I was fascinated by them—they were almost everywhere in the oceans. Eventually we learned to grow them in the lab. I was hooked!



Prochlorococcus cells and vesicles.

What has *Prochlorococcus* taught you?

We've known for a long time about diatoms and other relatively large photosynthetic organisms—phytoplankton—in the ocean. Those organisms were like the trees and the bushes. Suddenly, with the discovery of small photosynthetic bacteria such as *Prochlorococcus*, we learned, "Oh! There's grass, too!" Once we knew *Prochlorococcus* was there, we developed methods to monitor it in the field and sort out individual cells. We wondered how a single species could be so abundant in so many different areas of the ocean. We found out that this organism has extremely high levels of genetic diversity. This enables it to thrive in many different ocean environments, despite ever-changing conditions. Overall, studying *Prochlorococcus* has taught me to be humbled by the mind-blowing complexity of the natural world.

To date, what is the most surprising discovery from your lab?

The discovery of "grass" in the ocean was certainly a surprise! More recently, we've learned that the ocean is full of vesicles, very small lipid-bound particles that are budded off a cell. They look like little round peas on top of the cell. For years I had a picture of them that one of my students had taken, and I would ask people, "What do you think this is?" No one knew, until one of my post-docs speculated that they were vesicles, and

demonstrated this to be true. He discovered that they contain DNA, as well as RNA and a variety of proteins and lipids. We can determine which species produced a vesicle by sequencing the DNA it contains. We found out that many different species are producing vesicles. This is a beautiful example of how a casual observation in the lab—a photograph that I kept in a file cabinet for years—eventually led to the discovery of a new component of ocean ecosystems. Now the challenge is figuring out what they are doing!

> ***"Studying*Prochlorococcus*has taught me to be humbled by the mind-blowing complexity of the natural world."***

What do you find most rewarding about your job?

Working with *Prochlorococcus* is such a gift! Over the years, it's been an almost endless process of discovery—the organism is such a fundamental part of life in the ocean, and there are so many different questions you can ask. It has also been fruitful and exciting to work on *Prochlorococcus* with students and colleagues from many different backgrounds. It is wonderful to see my students working together to answer really hard questions. And I've learned so much from them—it has been a privilege and a joy to help them discover what they're best at and what they really enjoy doing. I love what I do. What could be more important than studying life?